Algebraic Number Theory: Study Notes

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Note

These notes provide a gradual introduction to algebraic number theory, beginning with the necessary algebraic foundations before developing the theory of number field. The exposition emphasizes clear definitions, rigorous proofs, and intuition, with the aim of building a solid understanding of the subject. This is a living document that will be continually revised, expanded, and refined as my study of algebraic number theory progresses, with new material, examples, and references incorporated over time.

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Chapter 1

Introduction to Commutative Algebra

1.1 Rings and Ideals

Commutative algebra begins with the study of rings and their ideals. These objects form the basic language for much of modern algebra, algebraic geometry, and number theory. In this chapter we introduce the fundamental definitions and properties that will be used throughout the text.

1.1.1 Rings and ring homomorphisms

A *ring* R is a set equipped with two binary operations, addition and multiplication, such that:

1. $(R, +)$ is an abelian group with identity element 0;
2. multiplication is associative and distributive over addition;
3. R has an identity element 1 such that $1x = x1 = x$ for all $x \in R$;
4. the ring is commutative if, in addition, $xy = yx$ for all $x, y \in R$;

A subset S of a ring R which is also a ring under same binary operations is defined as *subring* of the ring R denoted as $S < R$. Unless stated otherwise, every ring in this text is assumed to be commutative and to have identity. The case $1 = 0$ is allowed; then the ring consists of a single element and is called the *zero ring*.
A *ring homomorphism* is a map $f : R \to S$ such that

$$f(x + y) = f(x) + f(y), \qquad f(xy) = f(x)f(y), \qquad f(1) = 1.$$

A subset $A \subseteq R$ is called a *subring* if it is closed under addition and multiplication and contains 1.

1.1.2 Ideals and quotient rings

A *left/right ideal* I of a ring R is an additive subgroup of R such that

$$RI/IR \subseteq I.$$

Equivalently, for every $r \in R$ and $a \in I$ we have $ra/ar \in I$.

Definition 1.1. A non-empty subset of a ring R is called a (two-sided) ideal of R if it is both left and right ideal. *i.e.*,

1. $a, b \in I \implies a - b \in I$, and

2. $a \in I, r \in R \implies ra, ar \in I$.

Remark 1.2. For a commutative ring, left ideal, right ideal, and two-sided ideal coincide. Since, in this text, we will mostly consider commutative rings, we only talk about ideal in general by considering any of the descriptions above.

Notation 1.3. If I is an ideal of ring R, we denote it as

$$I \triangleleft R.$$

Definition 1.4. Let $I \triangleleft R$, define a relation on R as; for $a, b \in R$,a is related to b iff $a - b \in I$ and write

$$a \equiv b(modI).$$

Then this relation is an equivalence relation on R and class of a, $cl(a) = I + a =$ the coset of I by a in R.

1. $I + a = I + b \iff a - b \in I$ and

2. $I + a = I \iff a \in I$.

Then $R/I = \{I + a : a \in R\}$, the set of all cosets $I + a; a \in R$ is defined as quotient(factor) set of R by I.

If $I \triangleleft R$, then the quotient group R/I becomes a ring under the rules

$$(I + a) + (I + b) = I + (a + b)\, and$$

$$(I = a)(I + b) = I + ab.$$

This ring is called the *quotient(factor) ring* or *residue class ring* of R modulo I.

Remark 1.5. The subrings of R/I are of the form J/I where $I \subseteq J < R$. And the ideals of R/I are of the form J/I where J is an ideal of R containg I.

From above remark, we have the following fact:

Proposition 1.6. *There is a one-to-one order-preserving correspondence between the ideals of J of R containing I and the ideals $\bar{J}$ of R/I given by $J = \phi^{-1}(\bar{J})$.*

There is a natural surjective ring homomorphism

$$\phi : R \to R/I, \qquad r \mapsto r + I,$$

whose kernel is precisely I.

Definition 1.7. A non-zero element $x \in R$ is a *zero-divisor* if there exists $y \neq 0$ in R such that $xy = 0$. A ring with no zero-divisors is called an *integral domain.* For example, $\mathbb{Z}$ is an Integral Domain.

Definition 1.8. An element $x \in R$ is a *unit* if there exists $y \in R$ such that $xy = 1$. The set of all units of R forms a multiplicative group.

Definition 1.9. An element $x \in R$ is *nilpotent* if $x^n = 0$ for some $n > 0$.

Definition 1.10 (Field)**.** A commutative ring with unity in which $1 \neq 0$ is called as *field* if its every non-zero element is a unit.

Every field is an Integral Domain but not conversely as $\mathbb{Z}$ is an integral domain but not a field.

Remark 1.11. If$f : A \longrightarrow B$ is any ring homomorphism, the kernel of $f(= f^{-1}(0))$, denoted by $ker f$ is an ideal of A, and the image of $f(= f(A))$ is a subring C of B; and f induces a ring isomorphism $A/ker f \cong C$.

If I and J are ideals of R, then their *sum*, *intersection*, and *product* are defined by

$$I + J = \{x + y : x \in I,\ y \in J\},$$

$$I \cap J = \{x : x \in I \,\&\, x \in J\},$$

and

$$IJ = \left\{ \sum_k x_k y_k : x_k \in I,\ y_k \in J \right\}.$$

These operations satisfy the usual commutative and associative laws, and the product distributes over sum:

$$I(J + K) = IJ + IK.$$

Two ideals I and J are said to be *coprime(/comaximal)* if

$$I + J = R.$$

In this case,

$$I \cap J = IJ.$$

Two distince maximal ideals are always comaximal, since if $M \neq N$ are maximal ideals, then

$$M + N = R.$$

However, distinct prime ideals need not be comaximal. For example, in the ring $\mathbb{K}[x, y]$; $\mathbb{K}$ is a field, (x) and (x, y) but $(x) + (x, y) = (x, y) \neq R$.

Remark 1.12. Arbitrary intersection of ideals, product of two ideals, and sum of two ideals are again ideals of the same ring.

1.1.3 Classification of ideals

$P \triangleleft R$ is called *prime* if

$$xy \in P \implies x \in P \text{ or } y \in P.$$

$M \triangleleft R\, s.t\, M \neq R$, is called *maximal* if there is no ideal J such that

$$M \triangleleft J \triangleleft R.$$

i.e., if $M \subsetneq I \subsetneq R$; where $I \triangleleft M$, then either $M = I$ or $I = R$.

Remark 1.13. Below are few observations about prime and maximal ideals (Remember that "ring" means commutative ring with unity 1):

- P is prime if and only if R/P is an integral domain;
- M is maximal if and only if R/M is a field;
- Every maximal ideal is prime;
- In a finite ring, every prime ideal is maximal;

- the zero ideal (0) is prime iff R is an Integral Domain. That is, if R contains no zero divisors;
- $p\mathbb{Z}$ is both prime and maximal in $\mathbb{Z}$ for all prime p.

If $f : A \longrightarrow B$ is a ring homomorphism and Q is a prime ideal in B, then $f^{-1}(Q)$ is a prime ideal in A, for $A/f^{-1}(Q)$ is isomorphic to a subring of B/Q and hence has no zero-divisor $\neq 0$. But if M is a maximal ideal of B it is not necessarily true that $f^{-1}(M)$ is maximal in A; all we can say for sure is that it is prime. (Example: $A = \mathbb{Z}$, $B = \mathbb{Q}$, $M = (0)$.)

Definition 1.14. Let $S \subseteq R$; where R is any ring, then the smallest ideal I containing S is defined as the ideal generated by S. It is, symbolically, denoted by $I = (S)$. That is, if $S \subseteq J \lhd R$ then $I \subseteq J$.

$$i.e., (S) = \bigcap_{S \subseteq J \lhd R} J$$

Now, consider the following

$$I = (a) = ma + ra + sa : m \in \mathbb{Z}, r, s \in R.$$

If $m = 0$, $s = 0$ then

$$\forall\, r \in R, ra \in I = (a)$$
$$\implies Ra \subseteq (a);$$

And, if $m = 0$, $r = 0$ then

$$aR \subseteq (a);$$

If R is a commutative ring with unity then, $\forall r \in R, a \in Ra, aR \implies (a) \subseteq Ra, aR$

$$\implies (a) = Ra = aR$$

Definition 1.15. An ideal generated by a single element is called as *Principle ideal.* That is, I is a principle ideal iff $I = (a)$ for some $a \in R$.

If $U \lhd R$ such that it contains unity then $U = R$. If $I \lhd R$ such that it contains a unit of R then $I = R$. The ideal $\{0\} = (0)$ often denoted by 0 and $R = (1)$. We note the following observation:

Observation 1.16. *$r \in R$ is a unit iff $(r) = R = (1)$.*

Proposition 1.17. *Let R be a non-zero ring. Then the following are equivalent:*

(i) *R is a field.*

(ii) *The only ideals of R are 0 and (1).*

(iii) *Every ring homomorphism from R into a non-zero ring is injective.*

Proof. (i) $\Rightarrow$ (ii). Let $I \neq 0$ be an ideal of R. Then I contains a non-zero element x. Since R is a field, x is a unit. Hence

$$I \supseteq (x) = R,$$

and therefore $I = R$.

(ii) $\Rightarrow$ (iii). Let

$$\varphi : R \longrightarrow S$$

be a ring homomorphism, where $S \neq 0$. Since $\ker(\varphi)$ is an ideal of R, hypothesis (ii) implies that either

$$\ker(\varphi) = 0$$

or

$$\ker(\varphi) = R.$$

The latter is impossible because then $\varphi(1) = 0$, forcing S to be the zero ring. Hence

$$\ker(\varphi) = 0,$$

so φ is injective.

(iii) $\Rightarrow$ (i). Let $x \in R$ be a non-zero element. Suppose that x is not a unit. Then the principal ideal $< x >$ is a proper ideal of R, so the quotient ring

$$R/(x)$$

is non-zero. Let

$$\pi : R \longrightarrow R/ < x >$$

be the natural quotient homomorphism. By hypothesis, π is injective. Since

$$\ker(\pi) = (x),$$

it follows that

$$(x) = (0),$$

which implies $x = 0$, a contradiction. Therefore every non-zero element of R is a unit, and hence R is a field. □

Theorem 1.18. *Every ring $R \neq \{0\}$ has at least one maximal ideal.*

Proof. We use Zorn's Lemma. Let Σ be the set of all ideals $\neq (1)$ in R. Order Σ by inclusion. Σ is not empty, since $(0) \in \Sigma$. To, apply Zorn's lemma we must show that every chain in Σ has an upper bound in Σ; let then (I_α) be a chain of ideals in Σ. So that for each pair of indices α, β we have either $I_\alpha \subseteq I_\beta$ or $I_\beta \subseteq I_\alpha$. Let $I = \cup_\alpha I_\alpha$. Then I is an ideal: For if $a, b \in I$ and $r \in R$, then $\exists\, \alpha, \beta$ such that $a \in I_\alpha, b \in I_\beta$, we may assume $I_\alpha \subseteq I_\beta$ then $a, b \in I_\beta$ where $I_\beta \lhd R$ and hence $a - b \in I_\beta$ and $ra \in I_\beta$. Consequently, $a - b \in I$ and $ra \in I$.
Now, $1 \notin I$ because $1 \notin I_\alpha\ \forall\ \alpha$. Hence $I \in \Sigma$ and I is an upper bound of the chain. Hence, by Zorn's lemma, Σ has a maximal element. □

Corollary 1.19. *f $I \neq (1)$ is an ideal of R, there exists a maximal ideal of R containing a.*

Proof. Applying Theorem 1.18 on R/I, we obtain a maximal ideal $\bar{M} = M/I$ of R/I. Then, by Remark 1.5 and Proposition 1.6, the desired conclusion follows. □

Corollary 1.20. *Every non-unit of R is contained in a maximal ideal.*

Remark 1.21. There exist rings with exactly one maximal ideal, for example fields. A ring R with exactly one maximal ideal M is called a local ring. The field $K = R/M$ is called the residue field of R.

Proposition 1.22. *i) Let R be a ring and $M \neq (1)$ an ideal of R such that every $x \in R - M$ is a unit in R. Then R is a local ring and M its maximal ideal.*

ii) Let R be a ring and M a maximal ideal of R, such that every element of $1 + M$ (i.e., every $1 + x$, where $x \in M$) is a unit in R. Then R is a local ring.

Proof. i) Every ideal $\neq 1$ consists of non-units, hence is contained in M. Hence M is the only maximal ideal of R.

ii) Let $x \in R - M$. Since M is maximal, the ideal generated by x and M is (1), hence there existy $y \in R$ and $t \in M$ such that $xy + t = 1$; hence $xy = 1 - t$ belongs to $1 + M$ and therefore is a unit. Now use (i) to obtain to conclusion.

□

A ring with only a finite number of maximal ideals is called *semi-local*.

Example 1.23. Every ideal in $\mathbb{Z}$ is of the form (m) for some $m \geq 0$. The ideal (m) is prime iff $m = 0$ or a prime number. All the ideals (p), where p is a prime number, are maximal: $\mathbb{Z}/(p)$ is the field of p elements.

Definition 1.24. A *principal ideal domain* is an integral domain in which every ideal is principal.

In such a ring every non-zero prime ideal is maximal. For if $(x) \neq 0$ is a prime ideal and $(y) \supset (x)$, we have $x \in (y)$, say $x = yz$, so that $yz \in (x)$ and $y \notin (x)$, hence $z \in (x)$: say $z = tx$. Then $x = yz = ytx$, so that $yt = 1$ and therefore $(y) = (1)$.

1.1.4 Nilradical and Jacobsan Radical

Definition 1.25. The *radical* of an ideal I is

$$\sqrt{I} = \{x \in R : x^n \in I \text{ for some } n > 0\}.$$

It is again an ideal of R.
The radical of the zero ideal is called the *nilradical* and is the set of all nilpotent elements of R.

Definition 1.26. The *Jacobson radical* of R is the intersection of all maximal ideals of R. That is, if $\mathcal{M}$ is the set of all maximal ideals of R, then $J = \cap_{M \in \mathcal{M}} M$ is defined as Jacobson radical.